BANK LOAN REPORT DOCUMENT

Data | Dashboard Tableau

A. Bank Loan Report | Summary

The objective of this report is to offer insights regarding primary loan metrics and their temporal variations. This report will facilitate data-informed decision-making, enable the tracking of portfolio health, and highlight trends to guide our strategic lending approach

KPI's:

Total Loan Application

```
SELECT COUNT(id) AS total_application FROM bank_loan_data
```

	total_application
1	38576

Month-to-date (MTD) Loan Application

```
SELECT COUNT(id) AS total_application FROM bank_loan_data
WHERE MONTH(issue_date) = 12
```

	total_application
1	4314

Previous Month-to-date (PMTD) Loan Application

```
SELECT COUNT(id) AS total_application FROM bank_loan_data
WHERE MONTH(issue_date) = 11
```

	total_application
1	4035

Total Founded Amount

```sql
SELECT SUM(loan_amount) AS total_founded_amount FROM bank_loan_data
```

	total_founded_amount
1	435757075

Month-to-date (MTD) Total Founded Amount

```sql
SELECT SUM(loan_amount) AS total_founded_amount FROM bank_loan_data
WHERE MONTH(issue_date) = 12
```

	total_founded_amount
1	53981425

Previous Month-to-date (PMTD) Total Founded Amount

```sql
SELECT SUM(loan_amount) AS total_founded_amount FROM bank_loan_data
WHERE MONTH(issue_date) = 11
```

	total_founded_amount
1	47754825

Total Amount Received

```sql
SELECT SUM(total_payment) AS Total_amount_collected FROM bank_loan_data
```

	Total_amount_collected
1	473070933

Month-to-date (MTD) Total Amount Received

```sql
SELECT SUM(total_payment) AS Total_amount_collected FROM bank_loan_data
WHERE MONTH(issue_date) = 12
```

	Total_amount_collected
1	58074380

Previous Month-to-date (PMTD) Total Amount Received

```
SELECT SUM(total_payment) AS Total_amount_collected FROM bank_loan_data
WHERE MONTH(issue_date) = 11
```

	Total_amount_collected
1	50132030

Average Interest Rate

```
SELECT AVG(int_rate)*100 AS Avg_interest_rate FROM bank_loan_data
```

	Avg_interest_rate
1	114472,130340108

Month-to-date (MTD) Average Interest Rate

```
SELECT AVG(int_rate)*100 AS mtd_Avg_interest_rate FROM bank_loan_data
WHERE MONTH(issue_date) = 12
```

	mtd_Avg_interest_rate
1	115977,79323134

Previous Month-to-date (PMTD) Average Interest Rate

```
SELECT AVG(int_rate)*100 AS pmtd_Avg_interest_rate FROM bank_loan_data
WHERE MONTH(issue_date) = 11
```

	pmtd_Avg_interest_rate
1	110348,252788104

Average DTI (Debt-to-income)

```
SELECT AVG(dti)*100 AS avg_dti FROM bank_loan_data
```

	avg_dti
1	119141,069576939

Month-to-date (MTD) Average Interest DTI (Debt-to-income)

```
SELECT AVG(dti)*100 AS avg_dti FROM bank_loan_data
WHERE MONTH(issue_date) = 12
```

	avg_dti
1	122612,749188688

Previous Month-to-date (PMTD) Average DTI (Debt-to-income)

```
SELECT AVG(dti)*100 AS avg_dti FROM bank_loan_data
WHERE MONTH(issue_date) = 11
```

	avg_dti
1	118928,674101611

GOOD LOAN Vs. Bad Loan Issued

To evaluate the performance of lending activities and assess the quality of the loan portfolio, we need to create a comprehensive report that distinguishes between 'Good Loans' and 'Bad Loans' based on specific loan status criteria

GOOD LOAN ISSUED:

Good Loan Percentage

```
SELECT
    (COUNT(CASE WHEN loan_status = 'Fully Paid' OR loan_status = 'Current' THEN id END)* 100.0)/
    COUNT(id) AS good_loan_percentage
FROM bank_loan_data
```

	good_loan_percentage
1	86.175342181667

Good Loan Applications

```
SELECT COUNT(id) AS good_loan_appl FROM bank_loan_data
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

	good_loan_appl
1	33243

Good Loan Founded Amount

```
SELECT SUM(loan_amount) AS good_loan_founded_amount FROM bank_loan_data
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

	good_loan_founded_amount
1	370224850

Good Loan Amount Received

```
SELECT SUM(total_payment) AS good_loan_amount FROM bank_loan_data
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

	good_loan_amount
1	435786170

BAD LOAN ISSUED:

Bad Loan Percentage

```
SELECT
    (COUNT(CASE WHEN loan_status = 'Charged Off' THEN id END)* 100.0)/
    COUNT(id) AS bad_loan_percentage
FROM bank_loan_data
```

	bad_loan_percentage
1	13.824657818332

Bad Loan Applications

```
SELECT COUNT(id) AS bad_loan_appl FROM bank_loan_data
WHERE loan_status = 'Charged Off'
```

	bad_loan_appl
1	5333

Bad Loan Founded Amount

```
SELECT SUM(loan_amount) AS bad_loan_founded_amount FROM bank_loan_data
WHERE loan_status = 'Charged Off'
```

	bad_loan_founded_amount
1	65532225

Bad Loan Amount Received

```
SELECT SUM(total_payment) AS bad_loan_amount FROM bank_loan_data
WHERE loan_status = 'Charged Off'
```

	bad_loan_amount
1	37284763

LOAN STATUS:

To obtain a comprehensive overview of our lending operations and monitor loan performance, we have developed an 'Overview' report categorized by 'Loan Status.' This report will serve as a valuable tool for analyzing and understanding key indicators associated with various loan statuses

```
SELECT
    loan_status,
    COUNT(id) AS LoanCount,
    SUM(total_payment) AS Total_amount_received,
    SUM(loan_amount) AS Total_founded_amount,
    AVG(int_rate*100) AS Interest_rate,
    AVG(dti*100) AS DTI
FROM bank_loan_data
GROUP BY loan_status
```

	loan_status	LoanCount	Total_amount_received	Total_founded_amount	Interest_rate	DTI
1	Fully Paid	32145	411586256	351358350	110478,687198631	117875,047441282
2	Current	1098	24199914	18866500	147304,735883424	133435,245901639
3	Charged Off	5333	37284763	65532225	131783,030189387	123829,101818864

```
SELECT
    loan_status,
    SUM(total_payment) AS mtd_total_amount_received,
    SUM(loan_amount) AS mtd_total_founded_amount
FROM bank_loan_data
WHERE MONTH(issue_date) = 12
GROUP BY loan_status
```

	loan_status	mtd_total_amount_received	mtd_total_founded_amount
1	Fully Paid	47815851	41302025
2	Current	4934318	3946625
3	Charged Off	5324211	8732775

B. Bank Loan Report | Overview

MONTH:

```sql
SELECT
    MONTH(issue_date) AS month_number,
    DATENAME(MONTH, issue_date) AS month_name,
    COUNT(id) AS total_loan_appl,
    SUM(loan_amount) AS total_founded_amount,
    SUM(total_payment) AS total_amount_received
FROM bank_loan_data
GROUP BY MONTH(issue_date), DATENAME(MONTH, issue_date)
ORDER BY MONTH(issue_date)
```

	month_number	month_name	total_loan_appl	total_founded_amount	total_amount_received
1	1	January	2332	25031650	27578836
2	2	February	2279	24647825	27717745
3	3	March	2627	28875700	32264400
4	4	April	2755	29800800	32495533
5	5	May	2911	31738350	33750523
6	6	June	3184	34161475	36164533
7	7	July	3366	35813900	38827220
8	8	August	3441	38149600	42682218
9	9	September	3536	40907725	43983948
10	10	October	3796	44893800	49399567
11	11	November	4035	47754825	50132030
12	12	December	4314	53981425	58074380

STATE:

```sql
SELECT
    address_state AS State,
    COUNT(id) AS total_loan_appl,
    SUM(loan_amount) AS total_founded_amount,
    SUM(total_payment) AS total_amount_received
FROM bank_loan_data
GROUP BY address_state
ORDER BY address_state
```

	State	total_loan_appl	total_founded_amount	total_amount_received
1	AK	78	1031800	1108570
2	AL	432	4949225	5492272
3	AR	236	2529700	2777875
4	AZ	833	9206000	10041986
5	CA	6894	78484125	83901234
6	CO	770	8976000	9845810
7	CT	730	8435575	9357612
8	DC	214	2652350	2921854
9	DE	110	1138100	1269136
10	FL	2773	30046125	31601905
11	GA	1355	15480325	16728040
12	HI	170	1850525	2080184
13	IA	5	56450	64482
14	ID	6	59750	65329
15	IL	1486	17124225	18875941
16	IN	9	86225	85521
17	KS	260	2872325	3247394
18	KY	320	3504100	3792530
19	LA	426	4498900	5001160
20	MA	1310	15051000	16676279
21	MD	1027	11911400	12985170
22	ME	3	9200	10808
23	MI	685	7829900	8543660
24	MN	592	6302600	6750746
25	MO	660	7151175	7692732
26	MS	19	139125	149342
27	MT	79	829525	892047
28	NC	759	8787575	9534813
29	NE	5	31700	24542
30	NH	161	1917900	2101386

TERM:

```sql
SELECT
    term as Term,
    COUNT(id) AS total_loan_appl,
    SUM(loan_amount) AS total_founded_amount,
    SUM(total_payment) AS total_amount_received
FROM bank_loan_data
GROUP BY term
ORDER BY term
```

	Term	total_loan_appl	total_founded_amount	total_amount_received
1	36 months	28237	273041225	294709458
2	60 months	10339	162715850	178361475

EMPLOYEE LENGTH:

```sql
SELECT
    emp_length AS employee_length,
    COUNT(id) AS total_loan_appl,
    SUM(loan_amount) AS total_founded_amount,
    SUM(total_payment) AS total_amount_received
FROM bank_loan_data
GROUP BY emp_length
ORDER BY emp_length
```

	employee_length	total_loan_appl	total_founded_amount	total_amount_received
1	< 1 year	4575	44210625	47545011
2	1 year	3229	32883125	35498348
3	10+ years	8870	116115950	125871616
4	2 years	4382	44967975	49206961
5	3 years	4088	43937850	47551832
6	4 years	3428	37600375	40964850
7	5 years	3273	36973625	40397571
8	6 years	2228	25612650	27908658
9	7 years	1772	20811725	22584136
10	8 years	1476	17558950	19025777
11	9 years	1255	15084225	16516173

PURPOSE:

```sql
SELECT
    purpose,
    COUNT(id) AS total_loan_appl,
    SUM(loan_amount) AS total_founded_amount,
    SUM(total_payment) AS total_amount_received
FROM bank_loan_data
GROUP BY purpose
ORDER BY purpose
```

	purpose	total_loan_appl	total_founded_amount	total_amount_received
1	car	1497	10223575	11324914
2	credit card	4998	58885175	65214084
3	Debt consolidation	18214	232459675	253801871
4	educational	315	2161650	2248380
5	home improvement	2876	33350775	36380930
6	house	366	4824925	5185538
7	major purchase	2110	17251600	18676927
8	medical	667	5533225	5851372
9	moving	559	3748125	3999899
10	other	3824	31155750	33289676
11	renewable_energy	94	845750	898931
12	small business	1776	24123100	23814817
13	vacation	352	1967950	2116738
14	wedding	928	9225800	10266856

HOME OWNERSHIP:

```sql
SELECT
    home_ownership,
    COUNT(id) AS total_loan_appl,
    SUM(loan_amount) AS total_founded_amount,
    SUM(total_payment) AS total_amount_received
FROM bank_loan_data
GROUP BY home_ownership
ORDER BY home_ownership
```

	home_ownership	total_loan_appl	total_founded_amount	total_amount_received
1	MORTGAGE	17198	219329150	238474438
2	NONE	3	16800	19053
3	OTHER	98	1044975	1025257
4	OWN	2838	29597675	31729129
5	RENT	18439	185768475	201823056